

STATISTICAL ANALYSIS: DATA TYPES AND TOOLS

COMM50 Postgraduate Study & Research Methods

Research Methods Principles

- Un-biased
- Repeatable
- Objective
- Transparent

...as applied to
making a decision
about a hypothesis



Statistical Analysis



- Descriptive statistics

- ▣ Give a broad summary of the main characteristics of the data 'how it looks'

- Inferential statistics

- ▣ Investigates relationships and differences in data: informs decision-making
 - ▣ Allows modelling, what-if, and predictions to be made

Statistical Analysis



- Discover patterns & trends in data
 - ▣ Often by *sampling* a population
 - ▣ Larger sample = more representative
 - ▣ Compare data sets & find relationships
- Computer-based tools
 - ▣ MS Excel, Matlab, MathCAD, Maple, MiniTAB, SPSS, etc.
 - ▣ Learn how to use the stats feature of at least one

Data Types

□ Nominal

- ▣ Non-numeric data, such as categories or descriptors, can be counted
- ▣ Survey data, for example (nationality, favourite drink, ...)

□ Ordinal

- ▣ Non-numeric or numeric data, with a defined sequence
- ▣ Likert or Semantic Differential scales, for example



Data Types

□ Interval

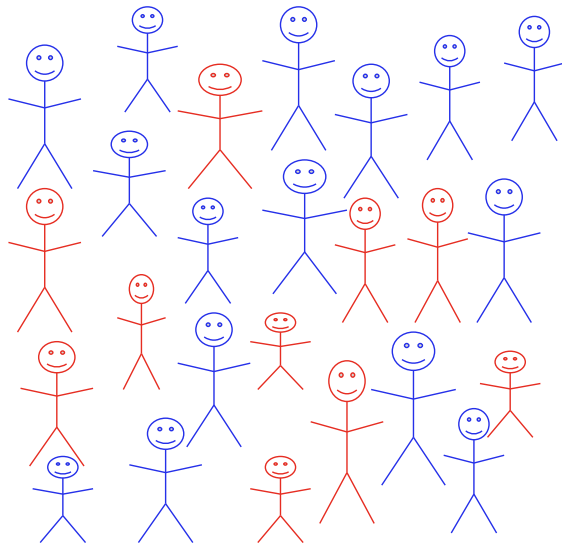
- ▣ Numeric defined scale with fixed distance between points
- ▣ Temperature ($^{\circ}\text{C}$ or $^{\circ}\text{F}$), for example

□ Ratio

- ▣ As interval, but where zero is not arbitrary, hence ratios meaningful
- ▣ Height, weight, or temperature in Kelvin, for example

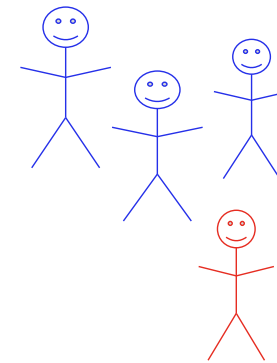


Sampling a Population



Population

Everybody that the research is targeted to be about



Sample

The subset of the population that actually participates in the research

Sampling a Population

- Probability sampling

- Simple random sampling
- Systematic sampling
- Stratified sampling

*Involves
randomness*

- Non-probability sampling

- Convenience sampling
- Quota sampling
- Snowball sampling

*Susceptible to
bias*

Sampling a Population

- Assuming a population of N participants:
- Random sampling
 - ▣ Chance of being selected $= 1/N$
- Systematic sampling
 - ▣ Select every z th person
- Stratified sampling
 - ▣ Identify existing population groups
 - ▣ Random sample ($1/N$ from each group)

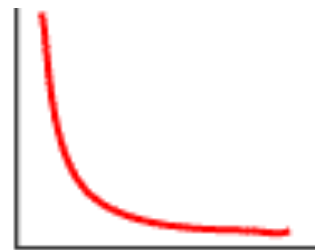
Sampling a Population



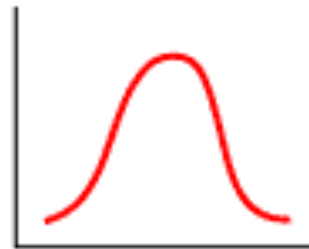
- Convenience sampling
 - ▣ Select participants who are easily obtainable
- Quota sampling
 - ▣ Identify existing population groups
 - ▣ Take from each group until target number reached
- Snowball sampling
 - ▣ After an initial set of participants are found they then invite participation from their contacts
 - ▣ Popular (and potentially effective) in the era of social media

Data Distributions

- A population is usually too large to quantify
 - ▣ Leads us to sample a population
- Frequency distribution indicates central tendencies and spread of values within a sample (vector)
- Histogram measures frequency of value occurrence in samples



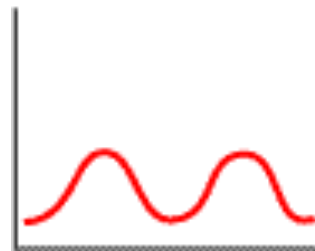
J-shaped



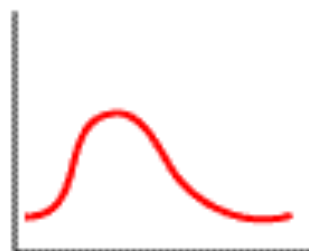
Normal



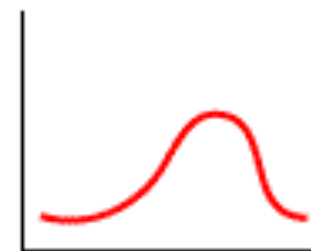
Rectangular



Bimodal

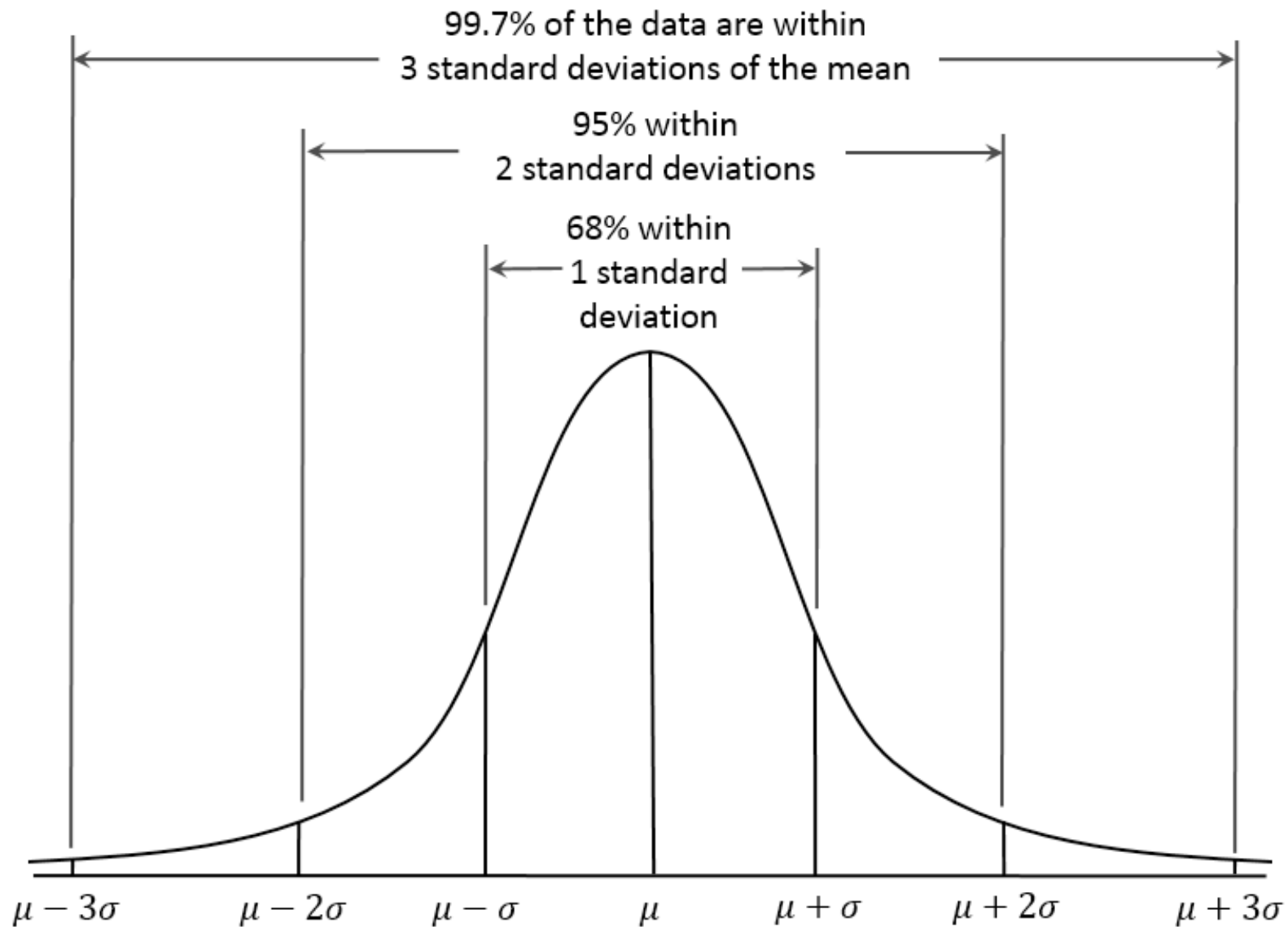


Positive (right) skew



Negative (left) skew

Normal Distribution

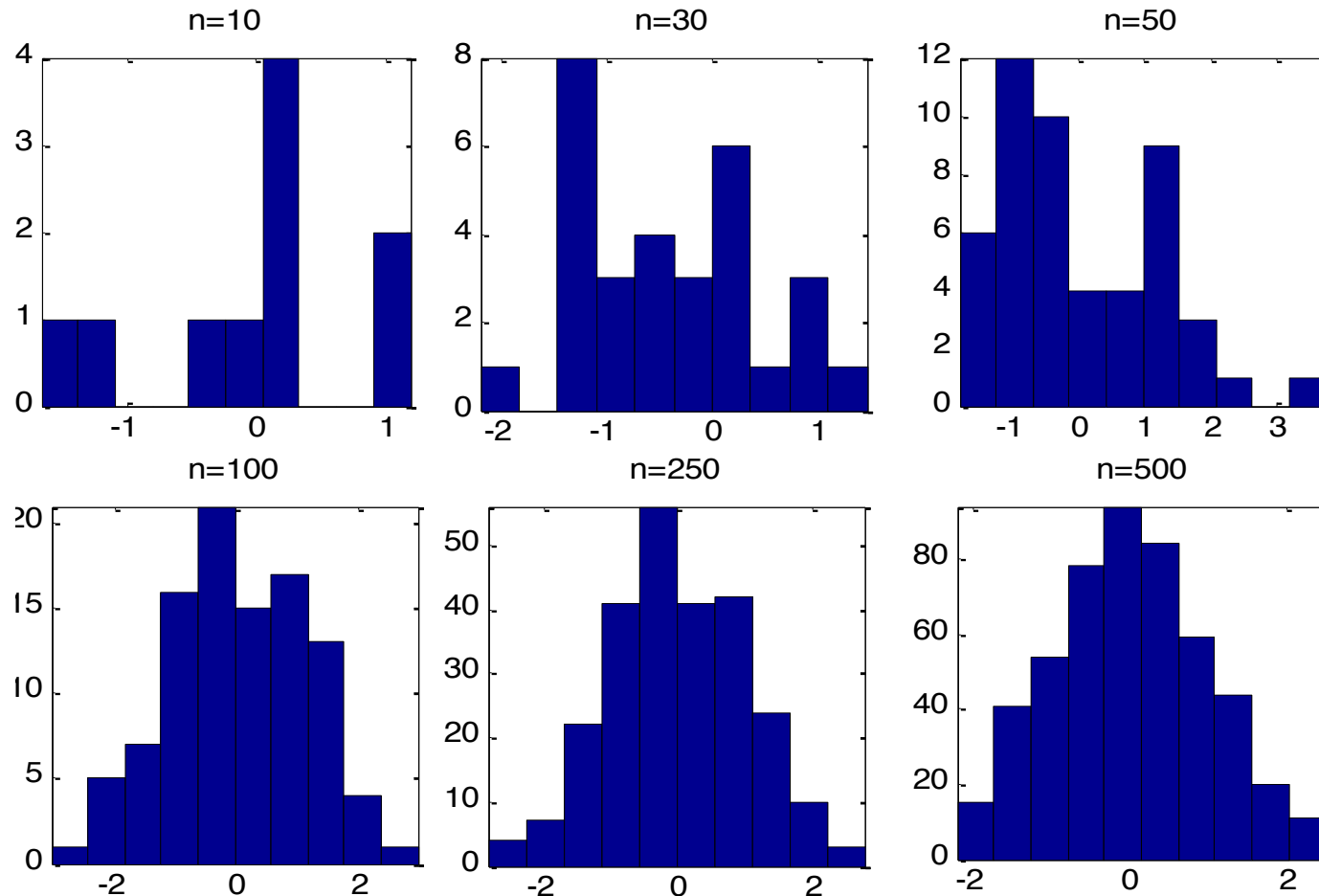


Sample Size and Sampling Error (noise)

- It's easier to detect a large effect with a small number of samples
- But subtle effects may be obscured by a small sample size
- Must also be pragmatic
 - ▣ What can we realistically do within a time frame?
 - ▣ What are the time and resource costs per sample and how big do we estimate the effect to be?
- Also consider
 - ▣ Dimensions of data being collected
 - ▣ Population size

Sample Size and Sampling Error (noise)

- More samples = more representative
- Small samples don't necessarily look like the population



Sample Size and Sampling Error (noise)

- The (model) population tells us there is a 50/50 chance of flipping a coin and seeing Heads or Tails
- Does this work in practice?
- How many Heads or Tails would you need to see before you decide the coin is rigged?



Sample Size Importance

- *Who is the better test match batsman based on average?*

Player	HS	Ave
AG Ganteaume (WI)	112	112.00
MN Waller (Zim)	72*	101.00
DG Bradman (Aus)	334	99.94
MN Nawaz (SL)	78*	99.00
VH Stollmeyer (WI)	96	96.00
DM Lewis (WI)	88	86.33
RE Redmond (NZ)	107	81.50
BA Richards (SA)	140	72.57
H Wood (Eng)	134*	68.00
Afaq Hussain (Pak)	35*	-



Sample Size Importance

- *More samples (in this case innings) indicates consistency & brings reliability*

Player	Mat	Inns	NO	Runs	HS	Ave
AG Ganteaume (WI)	1	1	0	112	112	112.00
MN Waller (Zim)	1	2	1	101	72*	101.00
DG Bradman (Aus)	52	80	10	6996	334	99.94
MN Nawaz (SL)	1	2	1	99	78*	99.00
VH Stollmeyer (WI)	1	1	0	96	96	96.00
DM Lewis (WI)	3	5	2	259	88	86.33
RE Redmond (NZ)	1	2	0	163	107	81.50
BA Richards (SA)	4	7	0	508	140	72.57
H Wood (Eng)	4	4	1	204	134*	68.00
Afaq Hussain (Pak)	2	4	4	66	35*	-

Discussion!



What statistics have you encountered before and in what situation?

Descriptive Statistics: the 3 'average' M's

□ Mean

- 'Average': Sum of all samples divided by number of samples

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

□ Median

- Middle value in a sample set that splits the distribution
- Mean of two middle values in even number set
- {1,3,10}; Median=3
- {1,3,5,7,9,10}; Median=6 (mean of two middle values)

□ Mode

- Most occurring actual data value in a sample set
- {4,3,7,3,6,7,3,3}; Mode=3
- {2,6,2,6,9}; Mode=2 & 6 (bi-modal)

Geometric Mean

- Useful in situations where values grow or where there are multiple variables of different magnitudes that contribute to a comparison
- Found by taking the n-th root of the product of samples

$$\overline{x}_{geo} = \left(\prod_{i=1}^n x_i \right)^{\frac{1}{n}}$$

Geometric Mean: Example

- Growth of number of users in a network (starting with 100 users):
 - ▣ Year 1 = 120 (20%)
 - ▣ Year 2 = 126 (5%)
 - ▣ Year 3 = 128[.52] (2%)
 - ▣ What is the mean growth year-on-year?
- Arithmetic mean = 1.09 (9%)
- Geometric mean = 1.087 (8.7%)

The Limitation of Means

- You need 31 runs to win the match...
 - ▣ Based on their mean performance in the last 12 months of ODI matches, which batsman would you ask to bat next?

Player	Inns	Runs	HS	Ave
PA Patel (India)	18	544	95	32.00
KP Pietersen (Eng)	18	571	78	31.72
AD Mathews (SL)	20	444	62	31.71
SR Patel (Eng)	8	190	70*	31.66
Tamim Iqbal (Ban)	21	633	95	31.65
EJG Morgan (Eng)	14	443	63	31.64
Kamran Akmal (Pak)	12	347	89	31.54

The Limitation of Means

- You need 31 runs to win the match...
 - ▣ Based on their mean performance in the last 12 months of ODI matches, which batsman would you ask to bat next?

<u>Player</u>	<u>Inns</u>	<u>Runs</u>	<u>HS</u>	<u>Ave</u>	<u>St. Dev</u>
PA Patel (India)	18	544	95	32.00	24.06
KP Pietersen (Eng)	18	571	78	31.72	21.70
AD Mathews (SL)	20	444	62	31.71	17.23
SR Patel (Eng)	8	190	70*	31.66	21.52
Tamim Iqbal (Ban)	21	633	95	31.65	27.43
EJG Morgan (Eng)	14	443	63	31.64	22.19
Kamran Akmal (Pak)	12	347	89	31.54	24.57

NB – These standard deviations do not differentiate between not out innings and out innings

Descriptive Statistics: Spread of Values

□ Range

- ▣ Distance between MIN & MAX values in a sample
- ▣ Range $\{5, 1, 3, 11, 7\} = 10$

□ Inter-Quartile Range (IQR)

- ▣ Distance between the upper quartile and lower quartile
- ▣ Or distance between the upper-half median and lower-half median
- ▣ For this set $\{2, 4, 6, 10, 14, 18, 20, 21, 23, 28, 32\}$
- ▣ Median = 18; IQR = 17 ($23 - 6$)

Descriptive Statistics: Spread of Values

- Variance

- ▣ Measurement of spread from mean

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

- Standard deviation (root-mean-square deviation)

- ▣ Returns spread from mean in same units as sample

$$\sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Comparing Variation Stats

- Measuring variation/deviation between groups is a common statistical analysis
- But the magnitude of the mean can contribute to the size of the deviation
 - ▣ Large means mean large deviations more likely
- The test or test scale may also vary between sample groups, meaning substantial differences in variation possible
 - ▣ e.g. measuring on a 5-point scale or a 15-point scale

Coefficient of Variation

- To counter this, we scale the standard deviation by the magnitude of the mean, producing the coefficient of variation (CV)

$$CV = \frac{st.dev}{mean}$$

- This produces a normalised measure of deviation between two sample sets
 - ▣ Often represented as a percentage ($CV * 100$)

CV Example

- Marks out of 100 in driving proficiency test using (say) 20 participants in each group
- It seems variability gets worse with experience
 - ▣ Instructors' st.dev more than 175% larger than learners(!)
- But, CV values very similar, indicating any difference is a result of the higher means:

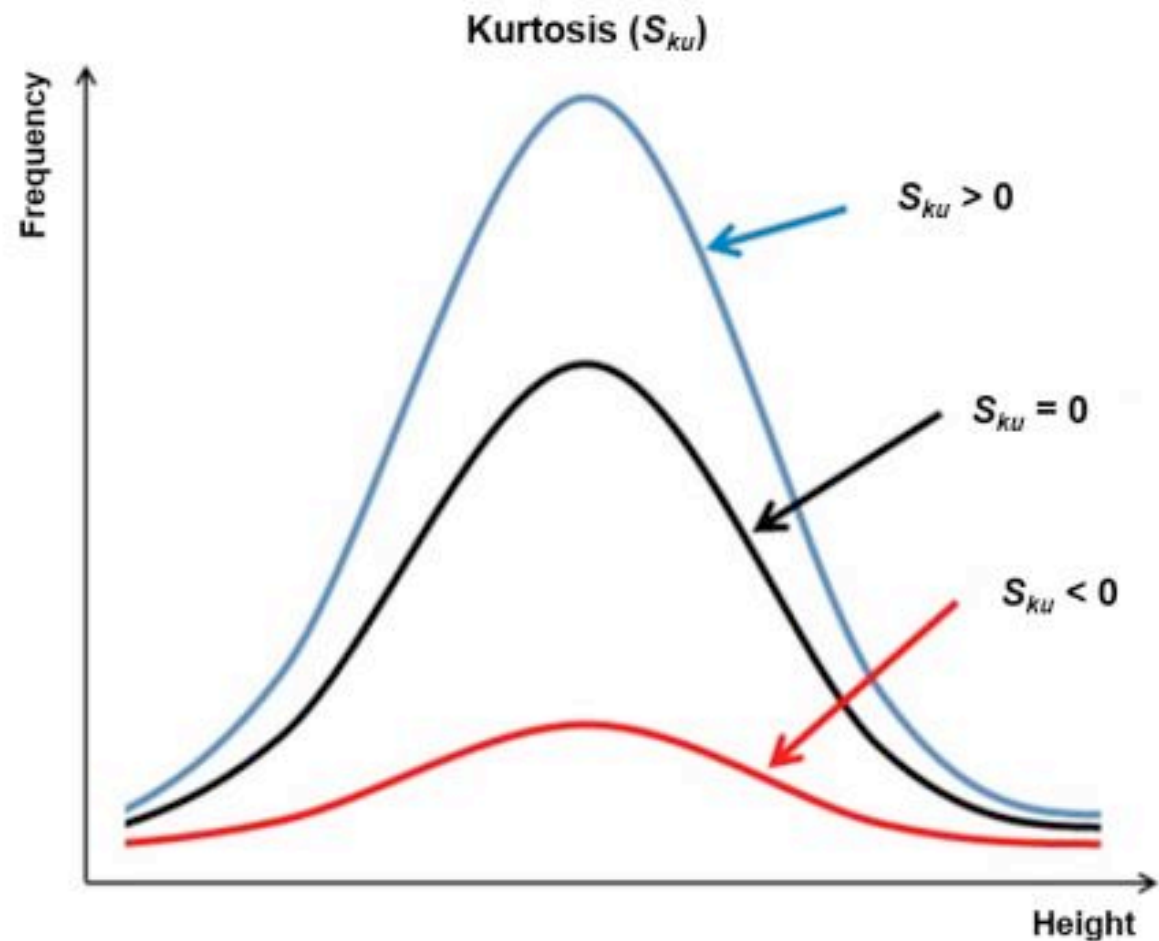
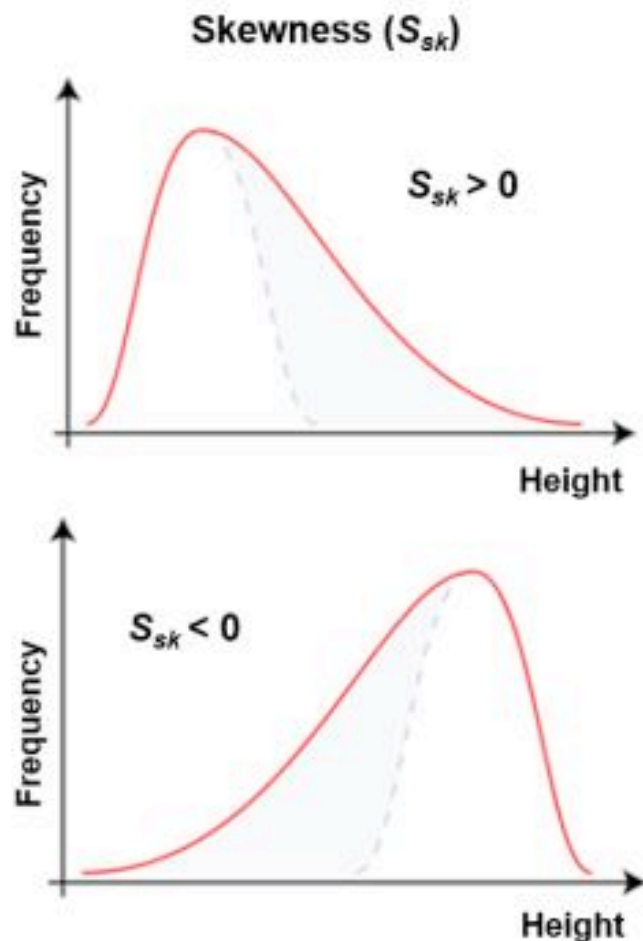
	Learner Drivers	Licensed Drivers	Driving Instructors
Mean	35.8	64.7	97.8
St.Dev	4.7	8.9	13
CV (%)	13.13	13.76	13.29

CV Example

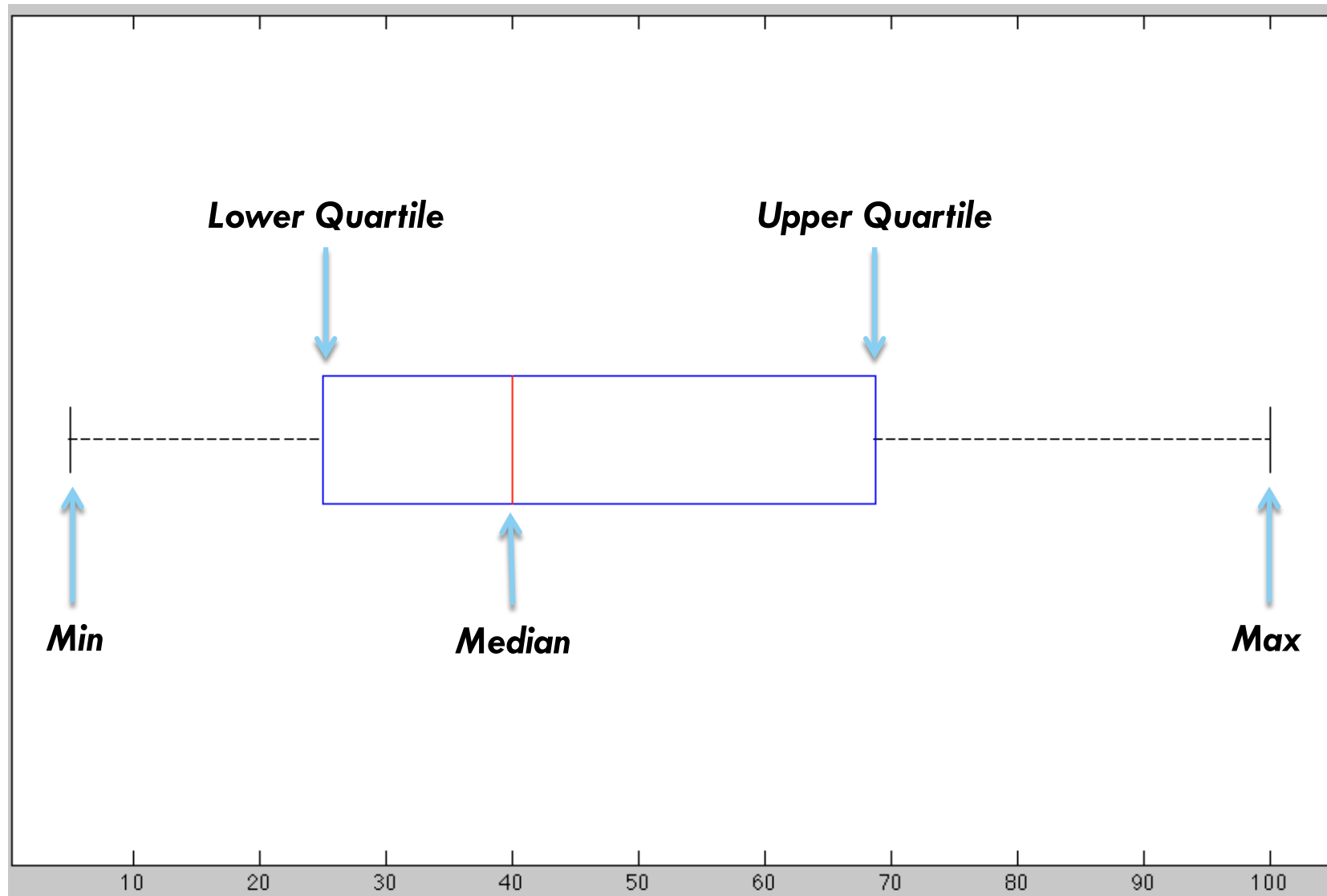
- Applying CV to previous examples in cricket:

<u>Player</u>	<u>Ave</u>	<u>St Dev</u>	<u>CV (%)</u>
EJG Morgan (Eng)	40.22	40.09	99.68
SCJ Broad (Eng)	39.83	32.33	81.18

Skewness and Kurtosis



Using Graphs to Show Distributions



Statistical Estimators

- Measures of central tendency and dispersion are key
- When choosing which to use, consider:
 - ▣ *Sufficiency*: a statistic that makes use of all of the data available in a sample
 - ▣ *Un-biased*: a statistic from a sample should be representative of the population
 - ▣ *Efficiency*: choose the statistic that best represents the population
 - ▣ *Resistance*: the statistic that is least influenced by outlying sample data

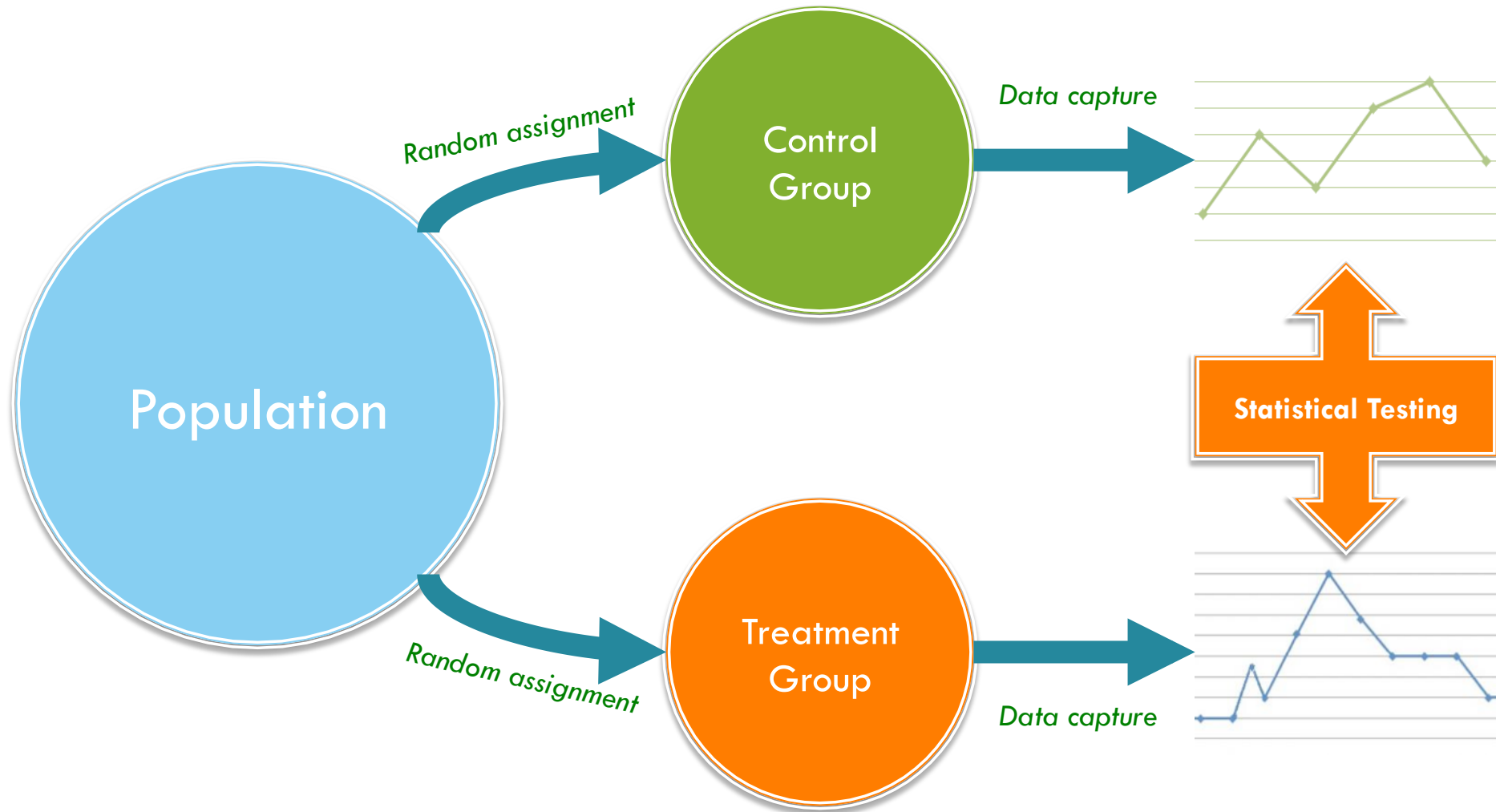
Randomized Controlled Trials (RCTs)

- Determine what is to be tested, in which population, and what it is being tested against
 - ▣ Control and Treatment(s)
- What outcome are you interested in and what will be the measure(s)?
- Randomly allocate samples to each group
 - ▣ Is the study blind or double-blind?
- After a defined period, compare the results from both groups
- Statistical analysis accounts for noise and provides objectivity

Randomized Controlled Trials (RCTs)

- Commonly encountered configurations in RCTs but also applicable in many other types of test
- Blind testing
 - ▣ Participants don't know if they are in the control or treatment group
- Double-blind testing
 - ▣ Participants don't know if they are in the control or treatment group
 - ▣ Researchers don't know which group is the control group and which is the treatment group

Randomized Controlled Trials (RCTs)



References



- Dahlberg, L., & McCaig, C. (Eds.). (2010). Practical research and evaluation: a start-to-finish guide for practitioners. Sage.
- Haynes, L., Goldacre, B., & Torgerson, D. (2012). Test, learn, adapt: developing public policy with randomised controlled trials. Cabinet Office-Behavioural Insights Team.